

NeuroCardia — Mobile App Design Spec

Reference document for the Flutter app UI (Module 3). Goal: a clinical-precision instrument feel, not a generic wellness-app look.

1. Design Principles

1. **The live waveform is the hero**, not a decorative icon of one. It is the largest, most prominent element on the home screen, drawn in real time as the beat happens.
 2. **One accent color, used consistently** to mean "this is live/active data." Everything else stays quiet — no competing bright colors.
 3. **No identical-card grid**. Avoid four matching rounded stat cards (heart rate / battery / sync / events) — this is the generic SaaS/wellness-app default. Use a single full-bleed waveform + a quiet status line + a log-style event list instead.
 4. **Numbers are tabular/monospaced** (BPM, confidence %, timestamps) — justified here because real ECG/monitor hardware uses fixed-width digit displays so values don't jitter as they update. This is one of the few domains where an "instrument" typographic choice is earned, not decorative.
 5. **Spend the "boldness budget" in one place**: the live waveform draw animation and R-peak marker. Keep transitions elsewhere minimal and purposeful (state changes, not decoration).
 6. **Empty/error states speak in the interface's voice**: state what happened and what to do, never vague, never apologetic.
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2. Color Tokens

Dark base, single accent — not the blue/teal wellness-gradient default.

Token	Hex	Use
bg.base	#0D0F12	App background
bg.surface	#15181C	Cards, sheets, nav bar
bg.surface-raised	#1C2025	Modals, elevated elements

Token	Hex	Use
text.primary	#EDEEF0	Primary text
text.secondary	#8A9099	Secondary/meta text
accent.signal	#4ADE80	Live/active data, normal beat, positive state
accent.alert	#E8A33D	Flagged/abnormal event (amber, not red — avoids false "emergency" alarm framing per Section 9 of the proposal, which explicitly excludes autonomous emergency alerting)
border.hairline	#2A2E33	Dividers, hairline rules only — no drop shadows

Avoid: soft blue-to-teal gradients, pastel wellness palettes, red for alerts (reads as "emergency," which overstates what this app does), generic grey card shadows (rgba(0,0,0,.1)) under every element.

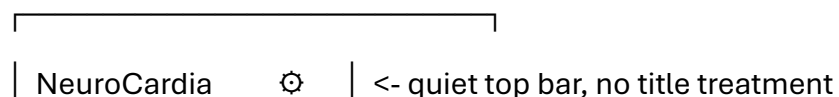
3. Typography

Role	Typeface	Notes
Numeric / data (BPM, confidence, timestamps, class labels)	A monospace face, e.g. JetBrains Mono or IBM Plex Mono	Tabular figures so digits don't shift width as they update live
Everything else (labels, body, nav)	A plain, restrained sans, e.g. Inter or IBM Plex Sans	Two families max, clearly distinct from each other

- Left-align text; don't center body content.
- No tracked-out ALL-CAPS eyebrow labels above sections.
- No accenting a single word in a headline with italic/bold/color.

4. Screen Layouts

4.1 Home / Live Monitoring Screen



```

|           |
| _____ |
| | ~~/\~/\~/\~/\~/\~/\~ | | <- full-bleed live waveform,
| | ●(R-peak marker) | | real-time draw-on animation
| | _____ | |
|           |
| 72 bpm · N · 98% confidence | <- single status line, mono type
|           |
| _____ | <- hairline rule, not a card border
| Recent events          |
| 10:42 V 91% conf.      | <- log/timeline style, not cards
| 09:15 N 99% conf.      |
| 08:03 S 87% conf.      |
| _____ |

```

- Waveform trace color: accent.signal (green) during normal rhythm; briefly shifts to accent.alert (amber) only on the flagged beat itself, then returns — not a persistent alarm-red banner.
- Status line and event list use the monospace numeric type.

4.2 Event Detail Screen

```

| _____ |
| ← Event · 10:42      |
|           |
| _____ |
| | [zoomed beat waveform] | | <- static snapshot of the flagged
| | _____ | | beat, not the live trace
|           |
| Classification V      |

```

Confidence	91%	<- mono, tabular, left-aligned
Recorded	10:42:07	
This is a research prototype		
and not a diagnosis.		
<- disclaimer in plain language,		
not legal boilerplate styling		

4.3 Offline / Sync States

- Offline: a single quiet line under the status text — "Recording locally · will sync when connected" — not a red banner or modal.
 - Syncing: subtle progress indicator inline, not a blocking spinner overlay.
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5. Motion

- One orchestrated moment: the live waveform draw + R-peak marker landing. This is the single place motion is spent.
 - No fade-and-slide-up entrances on list items, no hover/tap bounce on every element — these read as generic/AI-generated.
 - Sync and state-change transitions should be quick and functional (200ms), not decorative.
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6. Reference Material Reviewed

- Dark-mode healthcare UI kits (e.g. Moocare) — referenced for how a dark medical UI can read premium rather than cold; palette pushed further toward restraint than these kits typically use.
 - Real-time patient monitoring dashboards — referenced for live-feed + history split layout logic (applies to the Next.js dashboard too).
 - Heart-rate monitor UI kits (e.g. Pulsey) — referenced for live waveform-as-hero treatment; palette and card usage adapted away from their more decorative defaults.
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7. What This Rules Out

- Identical rounded stat-card grids
- Blue/teal wellness gradients
- Red alert banners (overstates the app's clinical role)
- Stock heartbeat-icon as logo/hero
- "Health score" big-number-with-small-label treatment
- Tracked-out ALL-CAPS section labels
- Decorative hover/entrance animation on every element